

Automated Black-box Prompt Engineering for Personalized Text-to-Image Generation

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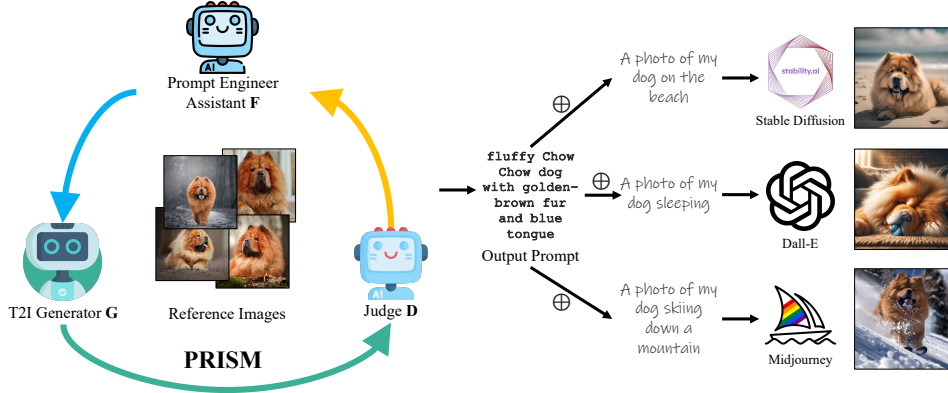


Figure 1: Given a set of reference images, our method, PRISM, is capable of creating human-interpretable and accurate prompts for the desired concept that are also transferable to both open-sourced and closed-sourced text-to-image models.

Recent advances in generative AI (GenAI) for art and image creation have sparked widespread interest, initiating dialogue between artists, content creators, and AI researchers. However, controlling the outputs of generative models remains a persistent challenge, particularly with traditional deep learning techniques. Early attempts, which often centered on particular architectures or tasks, were largely characterized by manually-curated data collection, fine-tuning, or retraining from scratch [1, 2, 3, 4]. These methods are not only resource-intensive but also struggle to generalize across different models and GenAI platforms, limiting their versatility in producing a diverse range of styles and ideas. As a result, despite the promise of these GenAI tools, there is a growing need for more efficient, adaptable, and accessible algorithms that enable better human control over AI-driven creativity.

Today, perhaps the most popular approach for controllable generation is to guide the generation process with a piece of textual information, or *prompt*, that describes the properties of the desired output using text-to-image (T2I) generative models [5, 6]. Through text, T2I models allow users to quickly and easily describe a wide variety of concepts, and users can more efficiently explore the behavior of the model through a myriad of strategies. The predominant method for obtaining such input text is to manually design candidate prompts in an iterative, trial-and-error fashion, a process known as *prompt engineering*, based on what the user (prompt engineer) *believes* will lead to a desirable output. Unfortunately, these practices are often time-consuming, sensitive to different phrasings [7], difficult to scale, and often requiring significant expertise to achieve desired results. Additionally, prompts that work for one platform may not generalize well to others, leading to inefficiencies and making the process mentally exhausting for users.

Motivated by the drawbacks of manual prompt engineering, a recent line of work known as *personalized* or *subject-driven* T2I generation has sought to automate the controllable generation pipeline. Given a collection of reference images that capture specific concepts, such as artistic style or shared objects, personalized T2I algorithms generate images that reflect those concepts. While personalized

T2I methods often involve fine-tuning or retraining the underlying T2I model [8, 9, 10], several approaches focus specifically on automating prompt engineering to generate effective prompts. However, these algorithms usually require pre-collected, architecture-specific keywords¹ or white-box, embedding-based optimization [11, 12], leading to non-interpretable prompts [13] and precluding the possibility of directly generating prompts for closed-source T2I models like Midjourney or DALL-E.

To address these shortcomings, we propose *Prompt Refinement and Iterative Sampling Mechanism* (PRISM), a new automated prompt engineering algorithm for personalized T2I generation. A key observation is that prompt engineers repeatedly update their “belief” of what makes an effective prompt based on the difference between their desired results and the generated images from previous iterations. Inspired by jailbreaking attacks on large language models (LLMs) [14], we design an algorithm that operates with only limited human input, is capable of generating human interpretable and editable prompts, makes minimal assumptions about the T2I generative model, and generalizes across different T2I platforms, including popular black-box models such as DALL-E and Midjourney.

Given a set of reference images, our method first generates an initial prompt and its corresponding image using a multimodal LLM and a T2I model. We then obtain a score indicating the visual similarity of the generated image and the reference image via another multimodal LLM. Leveraging LLMs’ in-context learning abilities [15, 16, 17], we instruct the LLM to update the candidate prompt distribution based on the previously generated prompt, images, and the evaluation scores. This processing is then repeated for a predetermined number of iterations. In the end, PRISM outputs the best-performing prompt by re-evaluating the top prompts generated from this process.

Through experiments, we demonstrate that PRISM consistently generates accurate, human-interpretable prompts for personalized T2I generation and direct image inversion (Figure 2), outperforming all baselines. With detailed experiments in the appendix, our method also shows significantly better generalizability and transferability as we achieve the best performance in almost all metrics when experimenting with closed-source models in comparison to existing methods. Finally, we show that because of the interpretability provided by our method, the prompts produced by PRISM are also easily editable (Figure 3), enabling a wide range of creativity possibilities in real life.



Figure 2: PRISM prompt for subject-driven T2I generation in various contexts and platforms.

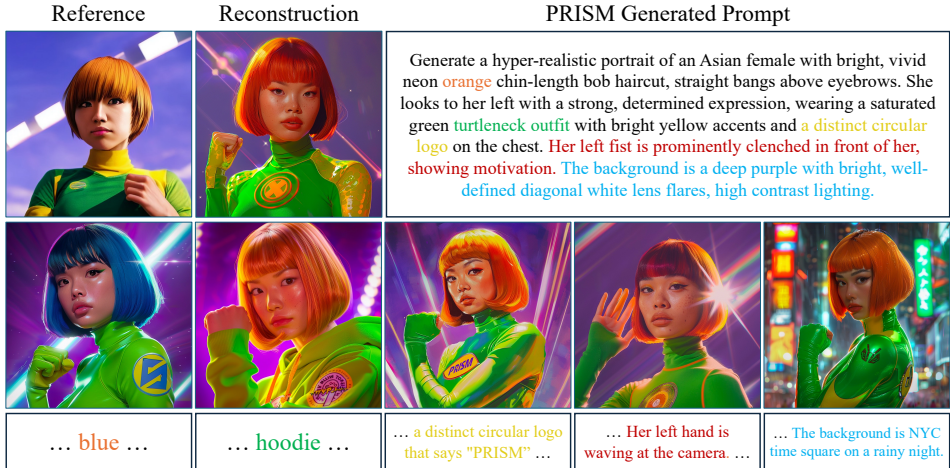


Figure 3: With accurate, human-interpretable direct image inversion from PRISM, artists can adjust specific attributes of a reference image with the rest of the scene intact on platforms like Midjourney.

¹<https://github.com/pharmapsychotic/clip-interrogator>

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